1.BANK MANAGEMENT SYSTEM

CODING:

```
from abc import ABC, abstractmethod

class Bank_Account(ABC):
    def __init__(self,name: str, balance: float):
        self.name = name
        self._balance = balance   # Encapsulation is applied


    def get_balance(self) -> float:
        return self._balance


    @abstractmethod
    def cal_interest(self) -> float:
        """Calculate interest in subclasses)."""
        pass


class Saving_Account(Bank_Account):
     # Applying 4% of interest
     interest_rate = 0.04
     def cal_interest(self) -> float:
         return self._balance * self.interest_rate

# Main program
print("/n**********Enter your details**********")
name = input("Enter your Name:")
balance = float(input("Enter your bank Balance:"))

account = Saving_Account(name, balance)
print(" ")
print(f"Your bank Balance: {account.get_balance()}")
print(f"Calculated Interest: {account.cal_interest()}")
```

OUTPUT:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                    Python Debug Console

PS C:\Users\hp\Desktop\ganga-py-training>  & 'c:\Program Files (x86)\Python314-32\python.exe' 'c:\Users\hp\.vscode\extensions\ms-
python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '52114' '--' 'C:\Users\hp\Desktop\ganga-py-training\bank_Acc.py
'
**********Enter your details**********
Enter your Name:Ganga
Enter your bank Balance:20000

Your bank Balance: 20000.0
Calculated Interest: 800.0
PS C:\Users\hp\Desktop\ganga-py-training>
```

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